TRACE-Fin: Evidence-Grounded Financial Investigation with Multi-Hop Retrieval, Temporal Risk Analysis, and Symbolic Verification

A Research Framework and Evaluation Protocol

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Abstract

Financial investigations often require analysts to connect evidence dispersed across narrative reports, tables, and time-ordered transaction records. A language model that produces a fluent answer is not sufficient: investigators also need verifiable source attribution, numerically correct calculations, sensitivity to rare suspicious behavior, and a clear indication of uncertainty. This paper proposes TRACE-Fin (Trustworthy Retrieval, Analysis, and Citation Evidence for Finance), a modular framework that combines (i) query decomposition and iterative document retrieval, (ii) temporal transaction-risk scoring with feature- and time-level attention, (iii) adaptive routing to symbolic operations for numerical claims, and (iv) claim-level evidence links with calibrated abstention. The design synthesizes multi-hop financial question answering, hierarchical temporal anomaly detection, and modular symbolic-neural reasoning, while adapting them to a single auditable investigation workflow.

Keywords: financial investigation; retrieval-augmented generation; multi-hop reasoning; anomaly detection; symbolic verification; evidence attribution; large language models

1 Introduction

Financial investigations are evidence-integration problems. An analyst may need to identify a transaction pattern, locate a related explanation in a report, combine values from a table, and explain how each source supports a conclusion. These tasks span different representations and time scales.

Recent language models and retrieval-augmented generation (RAG) systems can retrieve external material and synthesize answers, but grounding alone does not ensure that each claim follows from its cited evidence or that a calculation is correct [5]. For financial use, systems must distinguish a model-generated risk signal from verified evidence and expose uncertainty rather than convert incomplete information into an unqualified accusation.

This paper develops TRACE-Fin, a proposed framework for auditable financial investigation. Its design draws on three uploaded papers: Huang et al.s GAM-CoT Transformer [1]; Huang et al.s multi-hop LLaMA 3 RAG design [2]; and Huang et al.s MetaMath-LLaMA proposal [3].

2 Related Work and Design Foundations

2.1 Temporal transaction analysis: Blockchain AML research has studied both transaction-level classification and graph-based forensic methods. The Elliptic dataset provides time-indexed Bitcoin transaction data [4]. The uploaded GAM-CoT paper proposes a Global Attention Module (GAM) to reweight feature channels and time steps [1].

2.2 Financial RAG and multi-hop retrieval: RAG combines a language model with non-parametric retrieved evidence [5]. Financial QA benchmarks such as FinQA and TAT-QA test questions over financial reports, including numerical reasoning over text and tables [6, 7]. TRACE-Fin extends this pattern by tracking evidence coverage across hops and requiring every material answer claim to resolve to source spans.

2.3 Modular and symbolic reasoning: Chain-of-thought prompting established intermediate reasoning steps [8]. Tool-using approaches such as ReAct interleave reasoning and actions [9]. TRACE-Fin adapts these ideas to select among retrieval, temporal analysis, and deterministic calculation.

3 Problem Formulation

Let an investigation request be q. The available evidence consists of document collection D = {dj}, transaction sequences X =

{Xi}, and optional structured tables T = {Tk}. A system response is a set of answer claims A = {am} accompanied by a support ledger E = {(am, em, vm)}, where em identifies supporting evidence and vm records validation status.

The objective jointly optimizes answer utility, source support, calculation validity, and calibrated uncertainty: max_A [U(A, q) - lambda_u U_unsupported(A) - lambda_c C_invalid(A) - lambda_a C_abstain(A)].

4 TRACE-Fin Framework Architecture

4.1 Architecture overview: TRACE-Fin has five cooperating modules: an evidence normalizer, an investigation planner, a temporal risk encoder, a multi-hop retriever, and a verifier with a response composer. Evidence identifiers persist from ingestion through the final answer.

4.2 Evidence normalization and planning: Each ingested item receives a stable identifier and typed metadata: source name, location, timestamp, entity identifiers, and extraction confidence. The planner maps a query to typed subquestions Q and selects required operations O.

4.3 Temporal transaction-risk encoder: Gating mechanisms calculate global feature relevance and time-step relevance: gf = sigma(Wf meant(Hi,t) + bf), gt = sigma(Hi wt + bt), and composite gated representations with grouped temporal convolution.

4.4 Multi-hop retrieval: Candidate passages are ranked using a hybrid score s(q, d) = alpha s_lex(q, d) + (1-alpha) cos(q, d). Normalized context fusion weights beta_j = exp(s_j/tau)/Sigma exp(s_k/tau) guide multi-hop queries.

4.5 Symbolic verification: Claims are decomposed and typed as factual, numerical, temporal, relational, or model-derived. Support status vm in {supported, partially-supported, contradicted, unverified}.

5 Research Questions and Hypotheses

RQ1: Does feature- and time-aware temporal modeling improve illicit-class recall at a fixed false-positive rate over non-sequential baselines?

RQ2: Does iterative, coverage-guided retrieval improve evidence recall and citation support over one-hop retrieval?

RQ3: Does symbolic checking reduce arithmetic and unit errors relative to direct language-model generation?

RQ4: Does the integrated workflow improve supported-answer quality without imposing unacceptable latency or excessive abstention?

Hypotheses H1-H3 are preregistered for empirical verification with rigorous leakage control on Elliptic, FinQA, and TAT-QA benchmarks.

6 Discussion, Limitations and Conclusion

TRACE-Fin treats answer correctness, source support, numerical validity, risk ranking, and abstention as distinct outcomes. If validated, its modular architecture could help investigators move from an alert to a traceable explanation while preserving the difference between model-derived risk and source-established fact.

Conclusion: TRACE-Fin provides an evidence-grounded framework linking transaction sequences, financial documents, and symbolic verification into an auditable investigation workflow.

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